Lecture 1: Probability Review & Introduction to Statistical Inference

STAT 132

Contents

1 Review of Probability Theory **2**
- 1.1 Basic Concepts . . . 2
- 1.2 Random Variables . . . 2
- 1.3 Probability Distributions . . . 2
- 1.4 Expectation and Variance . . . 3
- 1.5 Important Distributions . . . 3
- 1.6 Law of Large Numbers and Central Limit Theorem . . . 4

2 Introduction to Statistical Inference **4**
- 2.1 What is Statistical Inference? . . . 4
- 2.2 Difference Between Probability and Inferential Statistics . . . 4
- 2.3 Example: Coin Tossing . . . 4

3 Parametric vs Non-Parametric Inference **5**
- 3.1 Parametric Inference . . . 5
- 3.2 Non-Parametric Inference . . . 5

4 Parameters and Parameter Space **5**
- 4.1 Parameters . . . 5
- 4.2 Parameter Space . . . 5

5 Statistics **5**
- 5.1 Example: Sample Mean and Variance . . . 6

6 Estimation and Prediction **6**
- 6.1 Estimation . . . 6
- 6.2 Prediction . . . 6

7 Hypothesis Testing **6**
- 7.1 Example: Testing a Mean . . . 6

1 Review of Probability Theory

1.1 Basic Concepts

- **Random Experiments**: An action or process that leads to one of several possible outcomes.
- **Sample Space (Ω)**: The set of all possible outcomes.
- **Events**: Subsets of the sample space.
- **Probability Measure ($\mathbb{P}$)**: A function assigning probabilities to events, satisfying:
  1. $0 \leq \mathbb{P}(A) \leq 1$ for any event A.
  2. $\mathbb{P}(\Omega) = 1$.
  3. For disjoint events A and B, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
- **Conditional probability**: The probability of event A given event B has occurred is denoted as $\mathbb{P}(A|B)$. If $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

- **Bayes rule**: Let A, B be two events. If $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)},$$

  where the denominator $\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|A^c)\mathbb{P}(A^c)$.

1.2 Random Variables

- A **Random Variable** is a function $X : \Omega \to \mathbb{R}$.
- **Discrete Random Variables**: Take on countable values.
- **Continuous Random Variables**: Take on values in an interval of $\mathbb{R}$.

1.3 Probability Distributions

- **Probability Mass Function (PMF)** for discrete variables: $f(x) = \mathbb{P}(X = x)$.
- **Probability Density Function (PDF)** for continuous variables: $f(x)$ such that $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)dx$.
- **Cumulative Distribution Function (CDF)**: $F(x) = P(X \leq x)$.

1.4 Expectation and Variance

- **Expectation (Mean)**:
  - Discrete: $E[X] = \sum_x x\mathbb{P}(X = x)$.
  - Continuous: $E[X] = \int_{-\infty}^{\infty} xf(x)dx$.
- **Variance**: $Var(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$.

1.5 Important Distributions

Name	Density/Mass Function	Mean	Variance	Parameter Restrictions
Binomial (Bin(n,p))	$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$	$n \in \mathbb{N}, 0 \leq p \leq 1$
Poisson (Poisson(λ))	$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ	$\lambda > 0$
Normal ($\mathcal{N}(\mu, \sigma^2)$)	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\sigma > 0$
Gamma (Gamma(α,β))	$f(x) = \frac{\beta^\alpha x^{\alpha-1} e^{-x\beta}}{\Gamma(\alpha)}$	α/β	α/β^2	$\alpha > 0, \beta > 0$
Beta (Beta(α,β))	$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\alpha > 0, \beta > 0$
Exponential (Exp(λ))	$f(x) = \lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\lambda > 0$
Uniform ($U(a,b)$)	$f(x) = \frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$a < b$

Table 1: Common Probability Distributions

Aside 1.1. The Gamma function, denoted as $\Gamma(x)$, is a mathematical function that extends the concept of factorials to non-integer and real numbers. You know:

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 1$$

for a positive integer n. The Gamma function does something similar but works for all positive real numbers, not just integers.

The Gamma function is defined as:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt, \quad x > 0.$$

While this might look complicated, you can think of it as a formula that calculates a value based on x, just like a factorial.

For positive integers n, the Gamma function is related to the factorial as follows:

$$\Gamma(n) = (n-1)!$$

For example:

$$\Gamma(1) = 0! = 1,$$
$$\Gamma(2) = 1! = 1,$$
$$\Gamma(3) = 2! = 2 \times 1 = 2.$$

In probability, the Gamma function appears in the probability density functions (PDFs) of distributions like the Gamma distribution.

The Beta function shows up in the PDF of the Beta distribution and is closely related to the Gamma function:

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x + y)}.$$

1.6 Law of Large Numbers and Central Limit Theorem

- **Law of Large Numbers**: As $n \to \infty$, the sample mean $\bar{X}$ converges to $E[X]$.
- **Central Limit Theorem**: For large n, the distribution of $\bar{X}$ approaches a normal distribution regardless of the original distribution of X.

2 Introduction to Statistical Inference

2.1 What is Statistical Inference?

Statistical inference involves making conclusions about populations using a sample of data drawn from the population. It bridges the gap between probability theory and real-world observations.

2.2 Difference Between Probability and Inferential Statistics

- **Probability Theory**: Starts with known parameters of a distribution and computes probabilities of events (data).
- **Inferential Statistics**: Starts with observed events (data) and makes inferences about the unknown parameters of the population distribution.

2.3 Example: Coin Tossing

Probability Problem: Given a fair coin (probability of heads $p = 0.5$), what is the probability of getting 6 heads in 10 tosses?

Inference Problem: You toss a coin 10 times and observe 8 heads. Can we infer whether the coin is fair? What is a good guess for the probability of heads for this coin?

3 Parametric vs Non-Parametric Inference

3.1 Parametric Inference

Assumes that the data follows a distribution characterized by a finite set of parameters.

Example: Assuming heights in a population are normally distributed with unknown mean μ and variance σ^2.

3.2 Non-Parametric Inference

Makes minimal assumptions about the form of the distribution.

Example: Estimating the median of a distribution without assuming a specific form.

4 Parameters and Parameter Space

4.1 Parameters

Quantities that index a family of probability distributions.

Example: In a normal distribution $N(\mu, \sigma^2)$, μ and σ^2 are parameters.

4.2 Parameter Space

The set of all possible values that parameters can take.

Example: For $N(\mu, \sigma^2)$, the parameter space is $\mu \in \mathbb{R}$, $\sigma^2 > 0$.

5 Statistics

Definition 5.1 (Statistic)**.** A **statistic** is a function of the data that does not depend on any unknown parameters.

$$T = t(X_1, X_2, \ldots, X_n)$$

5.1 Example: Sample Mean and Variance

- **Sample Mean**:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- **Sample Variance**:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

6 Estimation and Prediction

6.1 Estimation

- **Point Estimation**: Provides a single best guess of a parameter. We would like this point estimate to be "close" to the true (unknown) parameter value.
- **Interval Estimation**: Provides a range within which the parameter is expected to lie with a certain confidence level or probability.

6.2 Prediction

Using the model to predict future observations.

Example: Given the STAT 132 enrollment for the last 10 years, can we predict the enrollment for next year?

7 Hypothesis Testing

Hypothesis testing is a type of decision theory used to test a claim about a parameter using sample data.

7.1 Example: Testing a Mean

Problem: Is the average weight of a type of fruit equal to 100 grams?

- $H_0 : \mu = 100$
- $H_1 : \mu \neq 100$